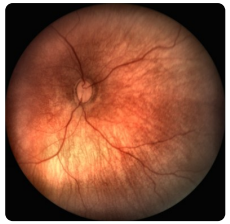
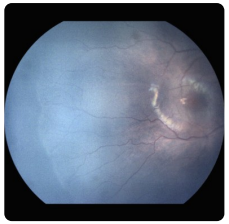
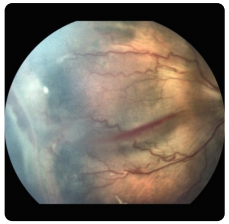
RetinAI final model: arm B, DINOv2 ViT-S/14

Six-class ROP staging from infant retinal photos. Selected by a rule fixed in advance and served in the app as 5 averaged models.

0.825	0.752	63.9%	0.937
CV QWK	New-hospital eye QWK	New-hospital eye accuracy	New-hospital any-ROP AUC

Example photo, as the model sees it						
	Normal	**Stage 1**	**Stage 2**	**Stage 3**	**Stage 4/5**	**AP-ROP**
Photos in the pool	3,837	577	959	620	89	618
Photos trained on	1,109	515	696	535	89	168
Cross-validation recall	90.9%	58.3%	56.1%	72.2%	88.2%	38.9%
Cross-validation precision	95.9%	30.2%	58.3%	78.8%	66.8%	70.5%
Cross-validation F1	0.933	0.398	0.572	0.753	0.759	0.501
New-hospital eyes	50	68	46	59	5	2
New-hospital recall, per eye	94.8%	51.0%	54.6%	56.6%	58.0%	70.0%
New-hospital precision, per eye	62.3%	60.7%	66.6%	86.7%	76.0%	8.8%
New-hospital F1, per eye	0.752	0.554	0.599	0.684	0.650	0.155

Model

Architecture	DINOv2 ViT-S/14 with registers (vision transformer)
Parameters	21.84 million per model
Pre-training	DINOv2, 142 million unlabelled everyday photos
Input	392 × 392 px, 28 × 28 patches of 14 px
Output	6 classes: Normal, Stage 1, 2, 3, 4/5, AP-ROP
Regularisation	Dropout 0.3, drop path 0.1
Preprocessing	Circle crop, letterbox, CLAHE (clip 2.0, 8 × 8), ImageNet normalise
Served as	5 fold models of seed 42, probabilities averaged
Stage	Top class of the averaged probabilities
ROP verdict	1 − P(Normal) at or above 0.9894
Per eye	The most severe photo decides

Training

Loss	Cross-entropy, class-balanced (β 0.999), label smoothing 0.05
Optimiser	AdamW, weight decay 0.05
Learning rate	Peak 1 × 10^{-4}, layer decay 0.8
Schedule	OneCycle: 10% warm-up, then cosine
Length	20 epochs, batch 16, mixed precision
Weights kept	Last epoch, no early stopping
Sampling	Natural order, no weighted sampler
Augmentation	Crop 60–100%, flips, ±15° rotation, colour, grey, resolution, affine, perspective, blur, erasing
Validation	5 folds split by infant, seeds 42 and 1337
Hardware	Kaggle Tesla T4, about 33 minutes per fold
Training time	16 runs, 9.0 hours for arm B

Accuracy

Test	Unit	QWK (95% CI)	Acc.	Macro-F1	AUC
Cross-validation	Photo	**0.825** (0.737–0.911)	76.6%	0.653	0.979
New hospital, served	Eye	**0.752**	63.9%	0.568	–
New hospital, single	Eye	**0.756** (0.679–0.821)	63.0%	0.566	0.937
New hospital, single	Photo	**0.416** (0.342–0.488)	43.0%	0.380	0.812
Unseen source, mean of 3	Photo	**0.834**	–	–	–

Screening, served model

Any-ROP eyes caught	65.6%
Stage 3 or worse caught	92.4%
Normal eyes not flagged	100.0%
ROP caught from one photo	31.4%
Calibration error (ECE)	0.144

Images

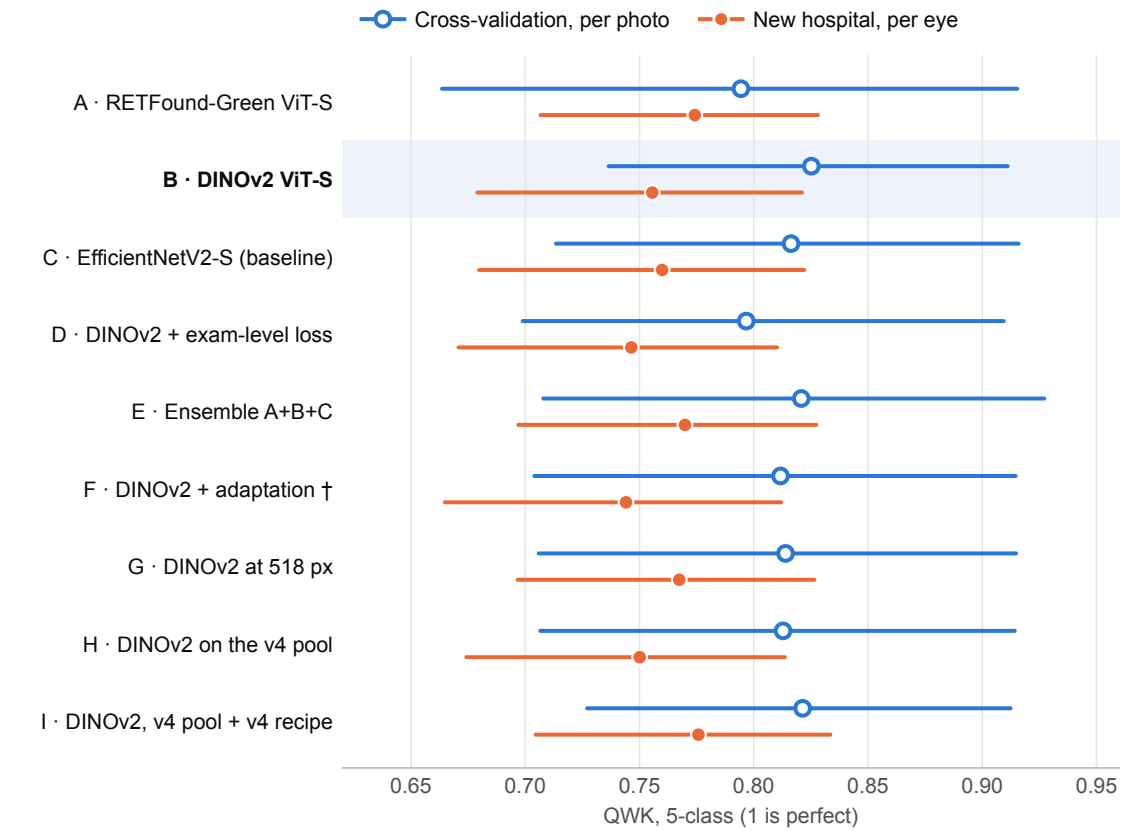
Collected	13,037 photos, 11 sources
Stage labelled	10,427
Used in the study	8,024 photos, 1,760 infants
Training pool	6,700 photos, 1,528 infants
Trained on	3,112
New hospital	663 dev + 661 sealed photos

Stage rows: arm B, both seeds; new-hospital rows average its 10 single models, on 230 eyes of the development half. "Served" is the five-model ensemble. New-hospital numbers are descriptive: the sealed half has not been scored for this model. Stage 4/5 and AP-ROP have too few new-hospital eyes to validate. Photos: Ostrava (CC0); ROP-VL and Shenzhen (CC BY 4.0, doi 10.6084/m9.figshare.30143461 and 10.6084/m9.figshare.25514449).

Nine models, no clear winner: B stays under the rule fixed in advance

The intervals overlap across all nine models. New-hospital scores played no part in the choice.

QWK with 95% intervals



Model	CV QWK	CV acc.	CV macro-F1	Eye QWK	Eye acc.	Eye macro-F1	Eye AUC
A · RETFound-Green ViT-S	0.794	75.8%	0.643	0.774	63.2%	0.563	0.935
B · DINOv2 ViT-S	**0.825**	**76.6%**	**0.653**	**0.756**	**63.0%**	**0.566**	**0.937**
C · EfficientNetV2-S (baseline)	0.816	78.8%	0.656	0.760	62.4%	0.573	0.943
D · DINOv2 + exam-level loss	0.797	80.2%	0.661	0.746	60.9%	0.542	0.935
E · Ensemble A+B+C	0.821	78.7%	0.664	0.770	63.9%	0.568	0.944
F · DINOv2 + adaptation	0.812	76.0%	0.644	0.744	62.4%	0.551	0.940
G · DINOv2 at 518 px	0.814	76.4%	0.647	0.767	62.6%	0.551	0.939
H · DINOv2 on the v4 pool	0.813	76.5%	0.646	0.750	62.0%	0.556	0.941
I · DINOv2, v4 pool + v4 recipe	0.821	75.0%	0.618	0.776	63.3%	0.560	0.944

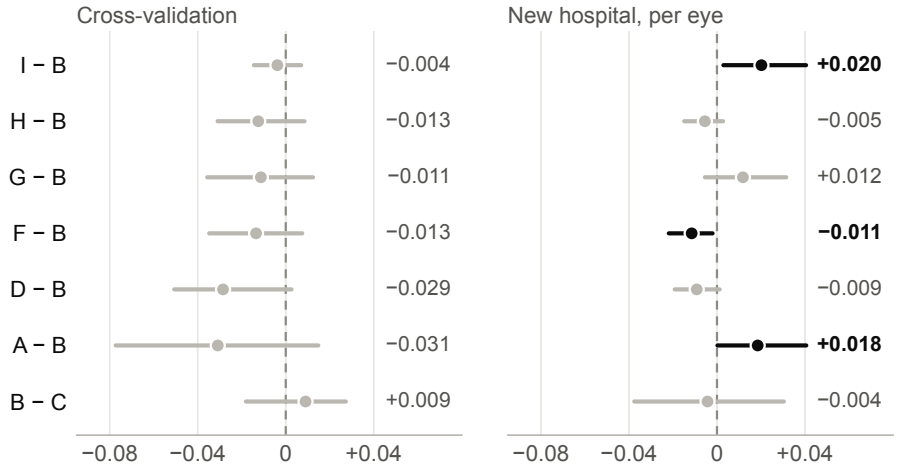
† F saw the new hospital's photos without labels, so its new-hospital numbers are not a clean test. "Eye" columns are the new hospital, per eye.

The v4 rule that kept B

Model	CV QWK	vs B	Needs	Mean LOSO	vs B	Needs	Outcome
I · DINOv2, v4 pool + v4 recipe	0.821	−0.004	≥ +0.010	0.839	+0.005	≥ −0.010	not adopted
H · DINOv2 on the v4 pool	0.813	−0.012	≥ +0.010	0.842	+0.009	≥ −0.010	not adopted
B · DINOv2 ViT-S	**0.825**			**0.834**			**stays**

A new model had to beat B on CV QWK by 0.010 and stay within 0.010 of B on the mean unseen-source score. Neither did.

Paired differences in QWK



Both models scored on the same resampled infants. Grey intervals cross zero. Dark intervals exclude it: I − B, F − B, A − B, all at the new hospital. New-hospital differences are descriptive and do not change the rule's outcome.

How H and I differ from B

Setting	B	H	I
Training rows	v3, 3,112	v4, 3,083	v4, 3,083
Label smoothing	0.05	0.05	0.1
Weight averaging (EMA)	off	off	0.999
Head warm-up	off	off	1 epoch
Rotation	±15°	±15°	±180°
Smallest crop	60%	60%	85%
Gamma, noise, Fourier swap	off	off	on

Stage by stage: Normal is strong, Stage 1 and AP-ROP are the weak spots

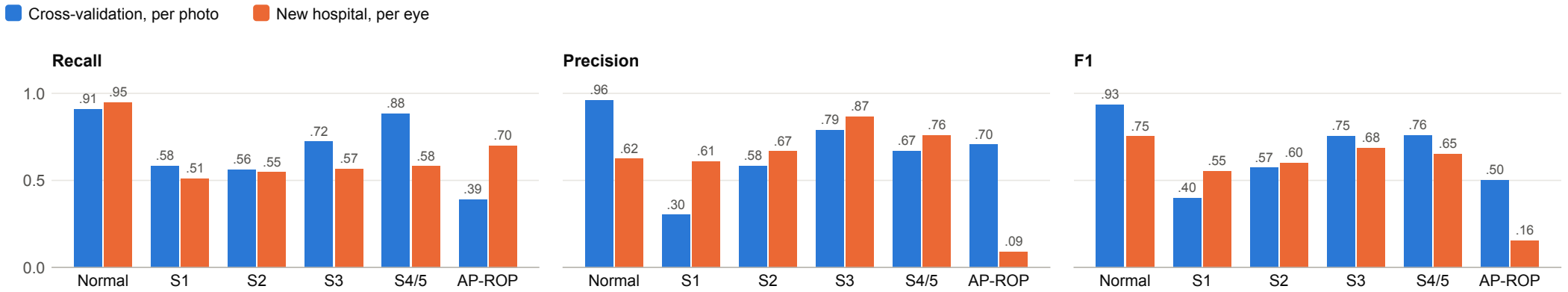
Rows are the true stage and columns the predicted stage. Each cell shows the count and its share of the row.

Confusion matrix, cross-validation, per photo

True stage \ Predicted stage	Normal	Stage 1	Stage 2	Stage 3	Stage 4/5	AP-ROP
Normal	**3,488** 91%	332 9%	10 0%	2 0%	1.5 0%	4 0%
Stage 1	71.5 12%	**334** 58%	144 25%	11.5 2%	0 0%	12.5 2%
Stage 2	50.5 5%	250 26%	**534** 56%	55.5 6%	7.5 1%	53.5 6%
Stage 3	11.5 2%	12 2%	111 18%	**446** 72%	9 1%	28 5%
Stage 4/5	1 1%	0 0%	0.5 1%	8 9%	**78.5** 88%	1 1%
AP-ROP	14 2%	181 29%	117 19%	43.5 7%	21.5 3%	**240** 39%

Confusion matrix, new hospital, per eye

True stage \ Predicted stage	Normal	Stage 1	Stage 2	Stage 3	Stage 4/5	AP-ROP
Normal	**47.4** 95%	2.6 5%	0 0%	0 0%	0 0%	0 0%
Stage 1	24.1 35%	**34.7** 51%	4.3 6%	2.3 3%	0 0%	2.6 4%
Stage 2	3.1 7%	11.2 24%	**25.1** 55%	2.3 5%	0 0%	4.3 9%
Stage 3	1.5 3%	7.9 13%	8.2 14%	**33.4** 57%	1 2%	7 12%
Stage 4/5	0 0%	1 20%	0.1 2%	0 0%	**2.9** 58%	1 20%
AP-ROP	0 0%	0 0%	0 0%	0.6 30%	0 0%	**1.4** 70%

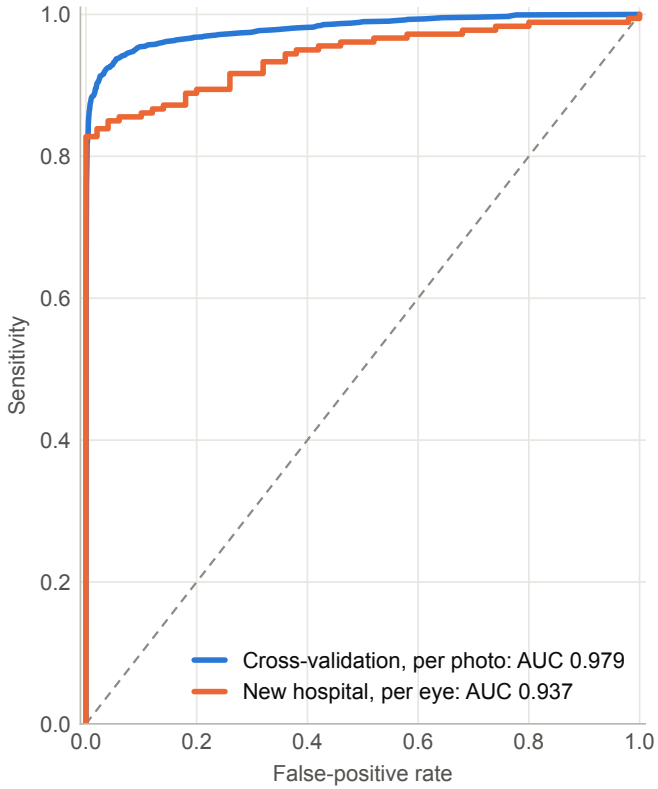


Counts are averaged over checkpoints, so new-hospital cells can be fractions. CV AP-ROP recall is low mainly because 470 of fold 0's 476 AP-ROP photos come from one infant, and many of those frames show no visible disease.

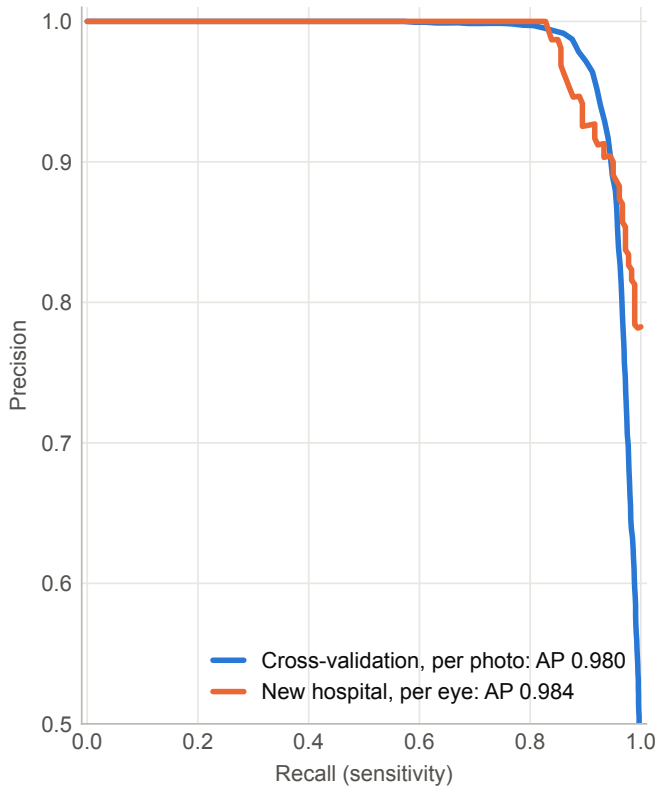
ROP against normal separates well, but the fixed line under-calls at the new hospital

The line was set on cross-validation to catch 90% of ROP. At the new hospital it caught 65.6% of ROP eyes.

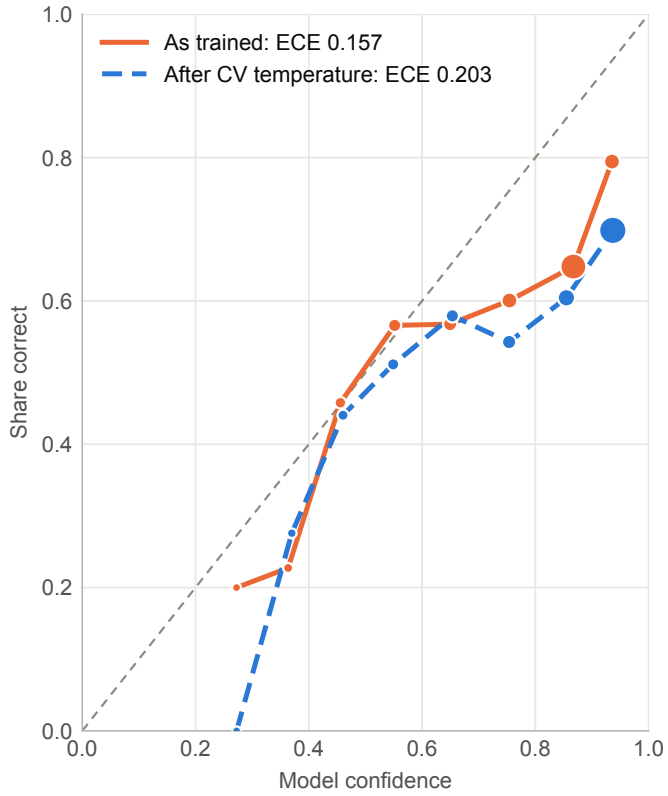
ROC curve



Precision–recall curve



Calibration, new hospital, per eye



Screening line

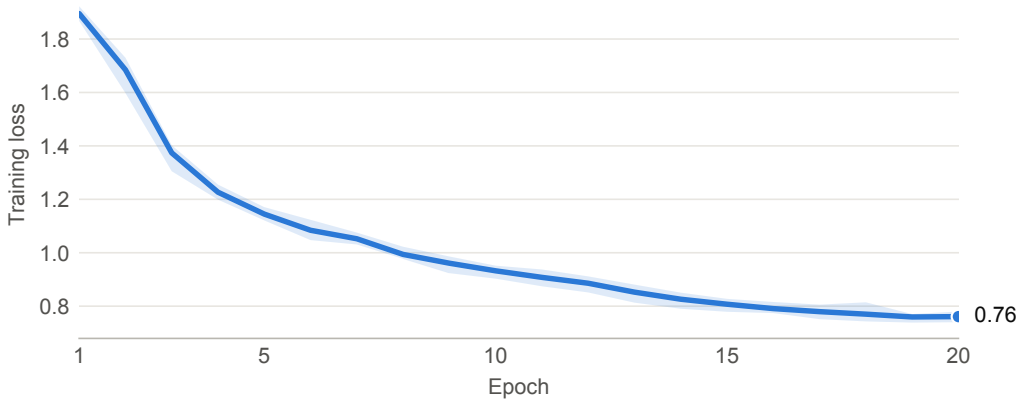
Model	Line	CV sensitivity	CV specificity	New-hospital sensitivity	New-hospital specificity	Stage 3 or worse caught
Served ensemble	**0.9894**	**–**	**–**	**65.6%**	**100.0%**	**92.4%**
Seed 42, single models	0.9894	90.0%	99.6%	65.6%	100.0%	92.7%
Seed 1337, single models	0.9893	90.0%	99.8%	67.6%	100.0%	94.2%

Score: 1 − P(Normal); per eye, the highest score among its photos. Curves use scores averaged over checkpoints; AUC and AP are means over checkpoints. The calibration chart shows single models; the served ensemble's ECE is 0.144. A negative result means "not flagged", not "healthy". Scores are not probabilities.

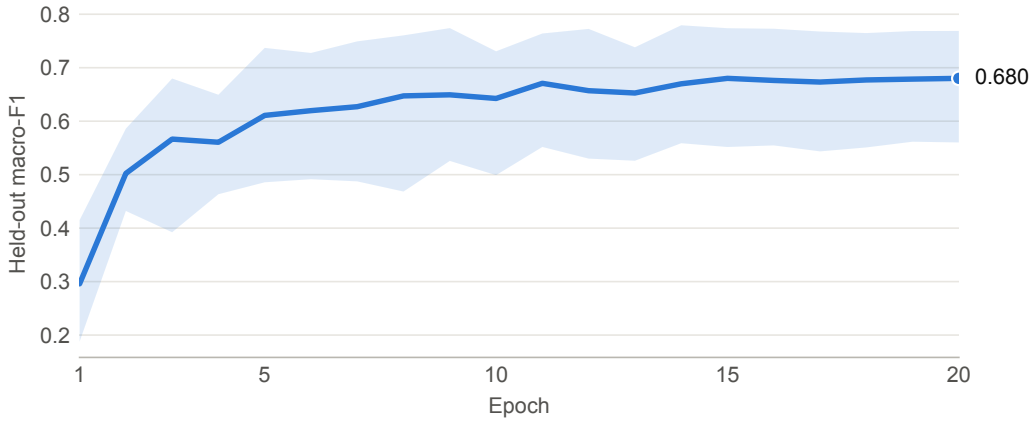
Training: 20 epochs on a Kaggle T4, about 33 minutes per fold

All logged runs: 92 runs, 60.0 hours. This is a minimum: June to August runs and arm C have no per-run times.

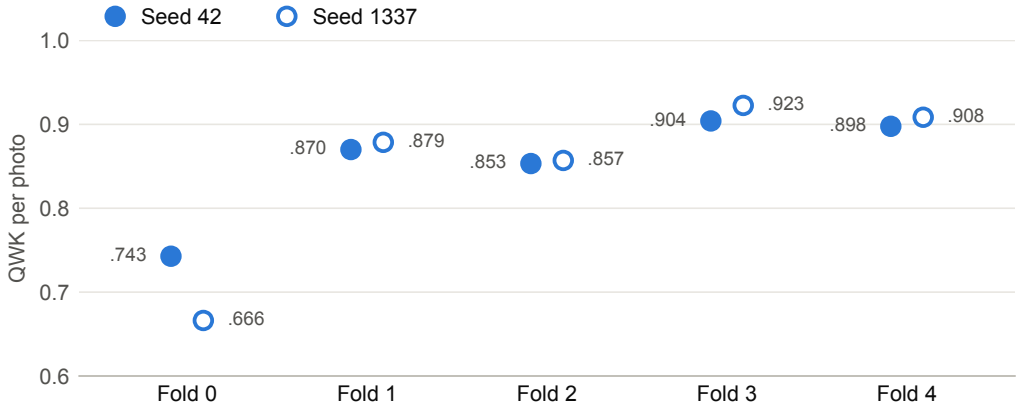
Training loss, arm B



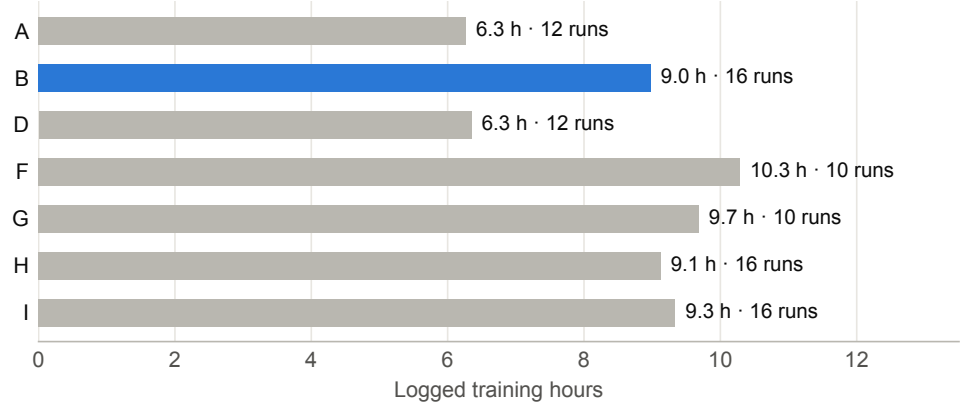
Held-out macro-F1, arm B



QWK on each fold, arm B



Training hours by model



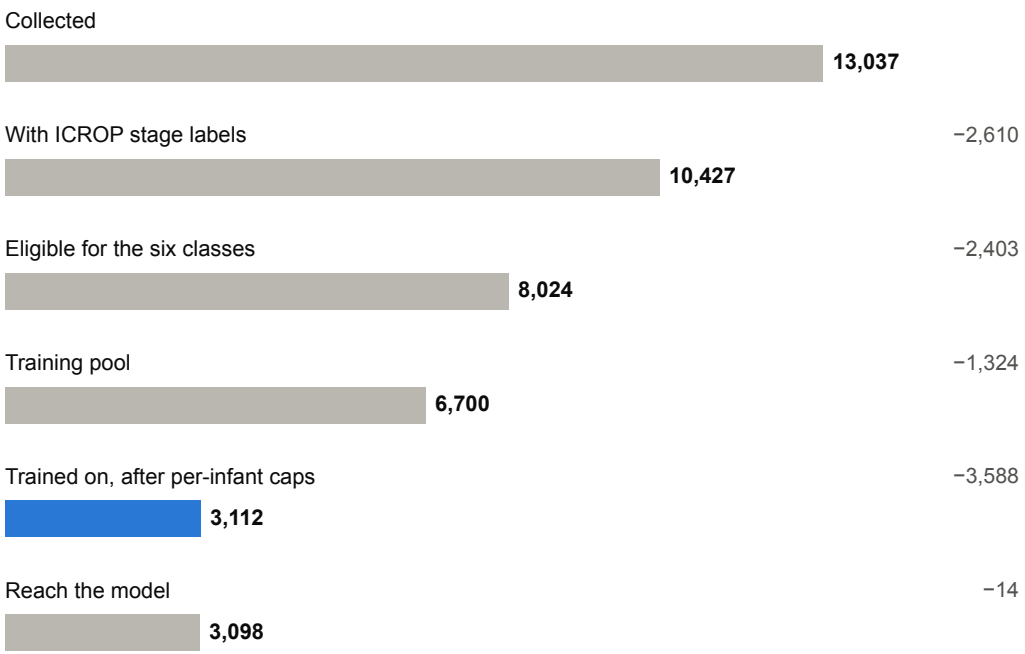
Fold	Train photos	Train infants	Validation photos	Minutes, seed 42	Minutes, seed 1337	QWK, seed 42	QWK, seed 1337
Fold 0	2,462	1,231	1,691	36.7	34.3	0.743	0.666
Fold 1	2,477	1,228	1,252	32.7	32.3	0.870	0.879
Fold 2	2,542	1,255	1,252	33.6	32.7	0.853	0.857
Fold 3	2,459	1,200	1,252	33.3	31.8	0.904	0.923
Fold 4	2,452	1,194	1,253	33.7	31.7	0.898	0.908

Line: mean of 10 runs. Band: range across runs. Fold 0 is lowest because it holds most AP-ROP photos.

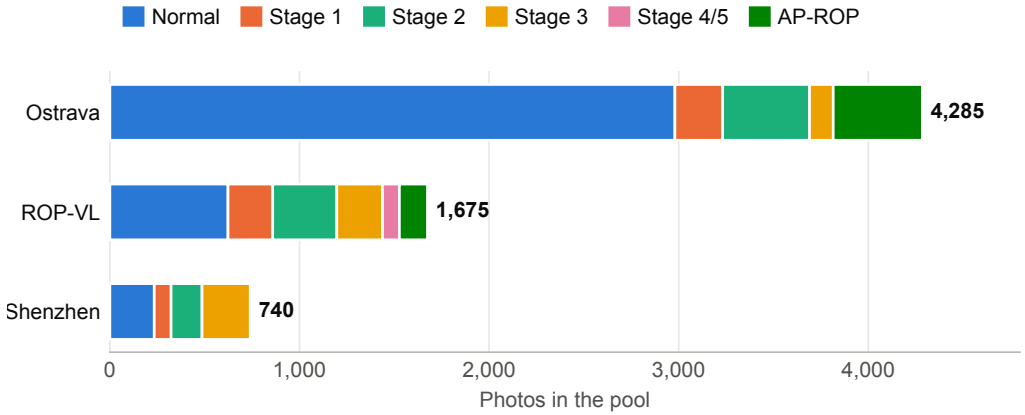
Data: 13,037 photos collected, 3,112 trained on

Preprocessing removes no photos. Every drop below comes from filtering or capping.

From collected photos to training photos



Training pool by source and class

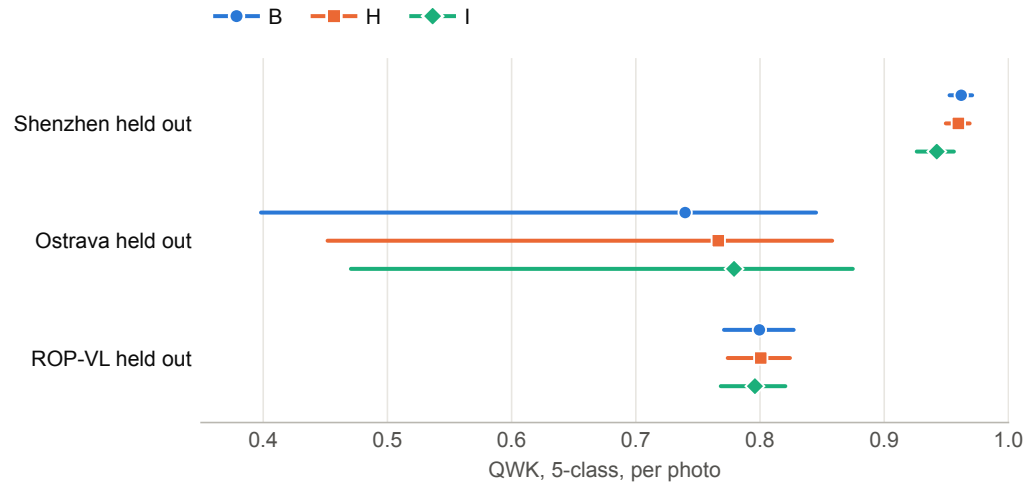


The ROP-VL fold trains with no Stage 4/5 photos, because only ROP-VL has them. ROP-VL and Shenzhen come from the same hospital but are different infants. Capped photos still count in validation.

The four image sources

Source	Country	Role	Collected	Used	Infants	Licence
Ostrava	Czech Republic	Train and validate	6,004	4,285	148	CC0
ROP-VL	China	Train and validate	2,020	1,675	983	CC BY 4.0
Shenzhen	China	Train and validate	1,099	740	397	CC BY 4.0
Multi-View	China	Test only	1,324	1,324	232	CC BY-NC 4.0

Whole source left out of training



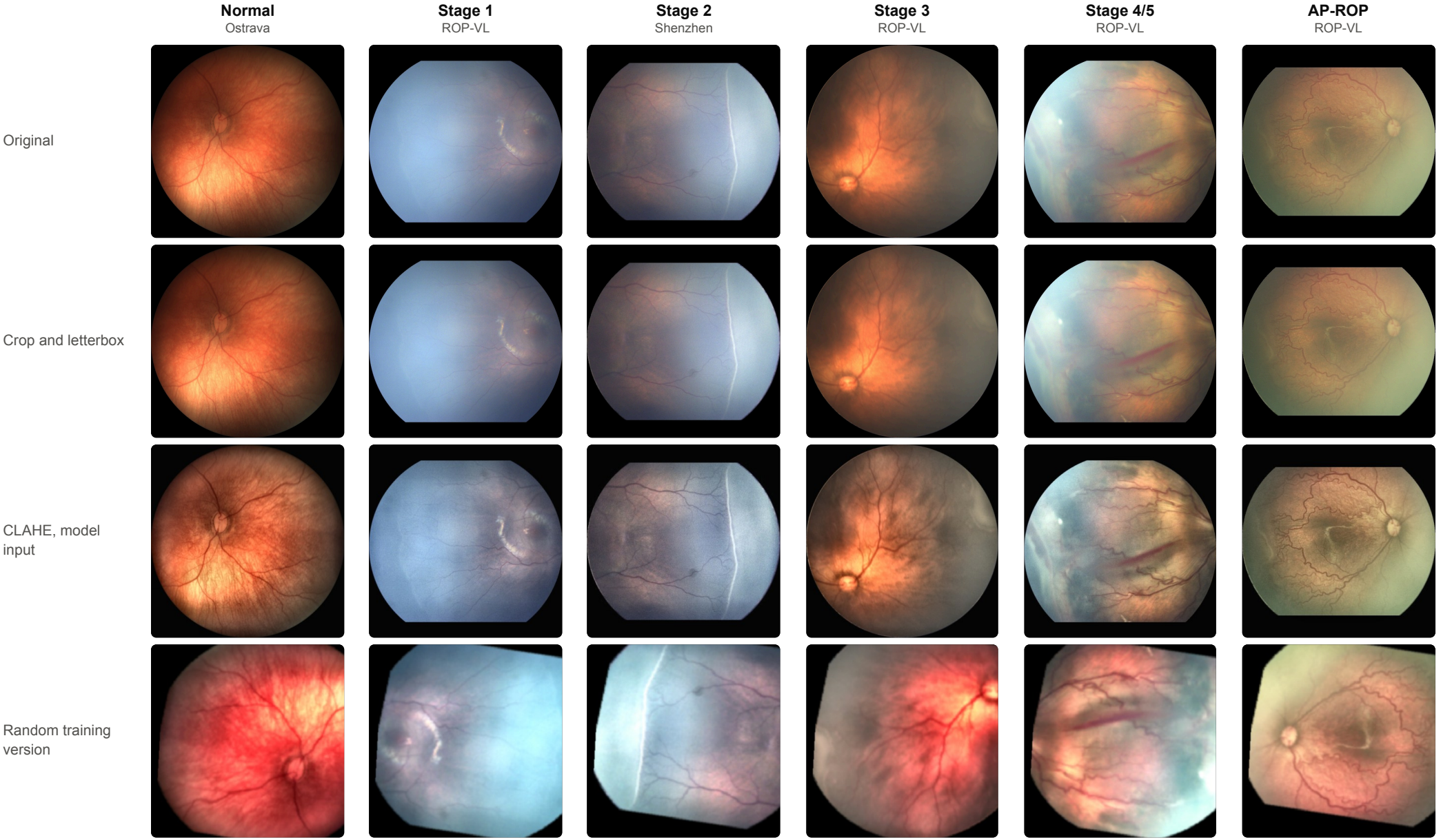
Model	Shenzhen	Ostrava	ROP-VL	Mean
B	**0.962**	**0.740**	**0.799**	**0.834**
H	0.960	0.766	0.801	0.842
I	0.942	0.779	0.796	0.839

Why photos left

Reason	Photos
Other datasets, no usable ICROP stage labels	2,590
Shenzhen copies of ROP-VL photos	20
Other disease, not ROP	1,295
Treated eyes	1,063
Stage 0	45
New hospital kept for testing	1,324
Above the per-infant caps	3,588
Exact duplicates	14

What the model sees: one real photo per class, through each step

The same code as the app. Examples only, not teaching images.

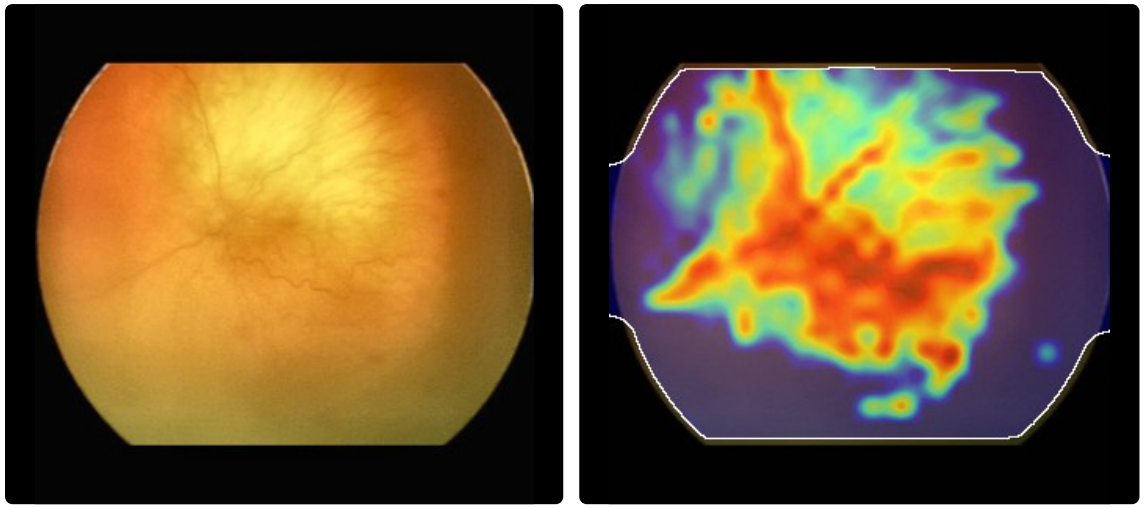


Labels are the datasets' own, and some photos show little visible disease. No new-hospital photo is shown. Sources: Ostrava (figshare collection 6626162, CC0); ROP-VL (doi 10.6084/m9.figshare.30143461, CC BY 4.0); Shenzhen (doi 10.6084/m9.figshare.25514449, CC BY 4.0).

What the app shows, and how to read the numbers

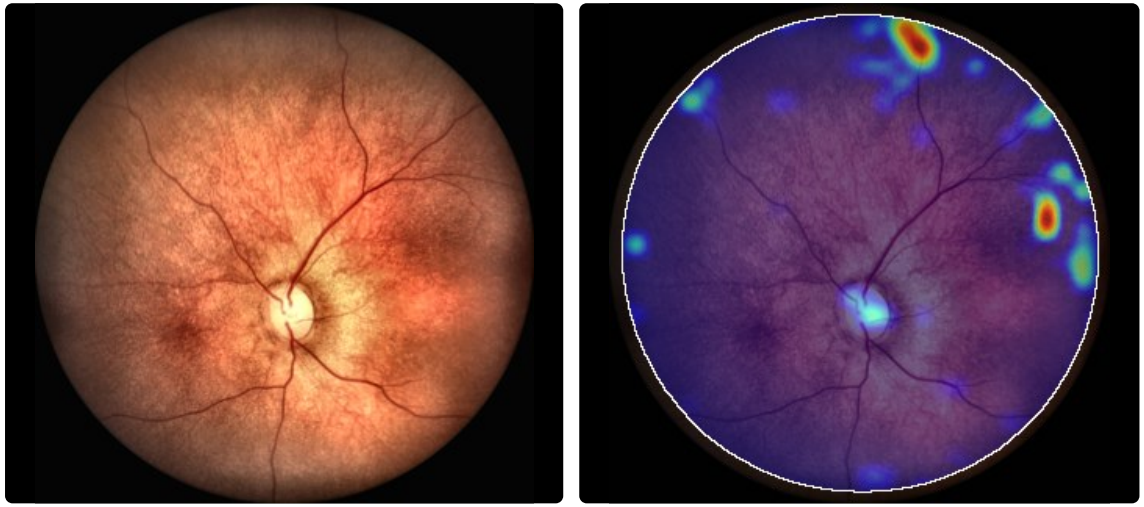
Heatmaps from the served five-model ensemble. Both infants are in the CV pool, so these show behaviour, not accuracy.

Photo with an AP-ROP label: model input and heatmap



ROP Detected. Stage AP-ROP, any-ROP score 99.1%. 0.6% of the heatmap weight falls off the retina.

Photo with a normal label: model input and heatmap



No ROP. Stage Normal, any-ROP score 11.5%. 85.6% of the heatmap weight falls off the retina.

The heatmap shows which regions moved the any-ROP score. It is not a lesion outline, and a bright area is not a diagnosis. Four of the five models trained on the AP-ROP photo, and on other photos of the normal infant.

Definitions

QWK	Agreement between predicted and true stage beyond chance. Big misses cost more. 1 is perfect, 0 is chance. "5-class" counts AP-ROP as Stage 4/5.
Accuracy	Share given exactly the right class.
Recall	Of the true cases, the share found. Also called sensitivity.
Precision	Of the cases called this class, the share that really are.
F1	Balances precision and recall. Macro-F1 averages it over classes equally.
AUC	Chance that a random ROP case scores above a random normal case.
ECE	Gap between confidence and how often the model is right.
CV	Cross-validation: tested on held-out infants from the three training sources.
LOSO	Leave one source out: trained on two sources, tested on the third.
Arm	One model variant, named A to I.

Arm B on each training source, cross-validation

Source	Scored per	QWK (95% CI)	Macro-F1
Ostrava	Session	0.701 (0.490–0.776)	0.424
ROP-VL	Eye exam	0.930 (0.915–0.943)	0.802
Shenzhen	Photo	0.967 (0.958–0.976)	0.865

Limits to state plainly

- A single photo caught only 31.4% of ROP photos, so use whole-eye mode.
- Stage 4/5 and AP-ROP are not externally validated.
- The nine models tie, so the limit looks like the labels, not the model.
- The sealed half is still to be scored once, for all nine models, after supervisor approval.

How the project got here, and what each choice cost

Scores from different rounds use different data and scoring rules, so compare them within a row, not across rows.

Every model generation

When	Model	Result	What happened
8 Jun 2026	EfficientNet-B3, one model for ROP and DR	ROP accuracy 73.3%, F1 0.71; DR macro-F1 0.50	Retired: split by photo, two diseases in one model
28 Jun 2026	EfficientNet-B0, EfficientNetV2-S, MobileNetV3-L, ResNet50, DenseNet121	ResNet50 best: AUC 0.964 validation, 0.927 test	ResNet50 deployed as the screener
Jun–Jul 2026	EfficientNetV2-S	Glaucoma AUC 0.973; DR AUC 0.906, QWK 0.728	Glaucoma withdrawn (duplicated G1020 photos); DR parked
2 Aug 2026	ResNet50, retrained by device	Camera shortcut confirmed: every ROP-positive test photo came from one camera	Retired 19 Aug: flagged 663 of 663 new-hospital photos, external AUC 0.691
7–9 Aug 2026	EfficientNetV2-S, DeiT3-S, ConvNeXt-T, CAFormer-S18, EfficientNet-B0	Macro-F1 0.492–0.520, inside run noise; cross-site 0.641 for EfficientNetV2-S	EfficientNetV2-S selected
Aug 2026	EfficientNetV2-S, structured head vs plain softmax	The two heads tie	Structured model served until 1 Nov
18 Oct 2026	Frozen RETFound-Green, DINOv2 and EfficientNet features with attention heads	Best frozen QWK 0.859 vs 0.917 fine-tuned; attention −0.080 QWK vs the most-severe-photo rule at the new hospital	Dropped; most-severe-photo rule kept
Oct 2026	v3: arms A–G	All close; B highest on cross-validation	B selected by the pre-registered rule
31 Oct–1 Nov 2026	v4: arms H and I	Neither beat B by the required +0.010 cross-validation QWK	B stays
1 Nov 2026	DINOv2 ViT-S/14, five fold models of seed 42	Replaces the EfficientNet model and its threshold	Current served model

Advantages and disadvantages

Choice	Advantage	Disadvantage
Circle crop	Removes the black border, so more of the frame is retina	It is a rectangular crop around non-black pixels, so one bright stray pixel can stop it trimming (from the code, not measured)
Letterbox	The retina keeps its true shape on every camera	Adds black padding: 24.7% of the frame on the August model
CLAHE	Improves local contrast; on RETFound-Green it added about 0.02 CV QWK and 0.10 new-hospital QWK	Never tested on DINOv2 alone, slow on CPU, and hospital colour differences remain
Saving preprocessed photos	Results barely change: QWK −0.0006, 99.2% same class (tested on one EfficientNet fold)	Must be rebuilt if preprocessing changes
392 px input	About 33 minutes per fold; 518 px took about 58 minutes and scored lower on CV (0.814 vs 0.825)	Less fine detail; 518 px was slightly higher on new-hospital eyes (0.767 vs 0.756), a tie
Augmentation	Makes the camera harder to recognise from colour and sharpness	No test of its effect alone was found
Split by infant	Honest scores, with no infant in both training and validation	Looks lower than papers that split by photo (one reports 99.7%)
Photo caps per infant	Without them, Ostrava's 10% of infants would give 64% of training photos	3,588 photos unused in training; AP-ROP drops from 618 to 168
Removing treated and other-disease photos	Clean six classes	Such an eye must still be forced into one of the six classes
New hospital kept out of training	A real test on an unseen hospital	Only 7 eyes of Stage 4/5 and AP-ROP in the dev half
DINOv2 ViT-S	Small (about 22M parameters) with the highest CV score	Not a clear win: the arms tie, and arm I scored higher at the new hospital
Five-model ensemble	Stable, and every fold contributes	Five times the compute with no accuracy gain (0.752, single models 0.745–0.774)
Last-epoch weights	Avoids picking a lucky epoch	May not be the best epoch
Most severe photo decides the eye	QWK 0.756 per eye vs 0.416 per photo	One wrong photo can over-grade an eye; AP-ROP precision is 0.09 (only 2 true AP-ROP eyes)
Screening line	No false alarms on 50 normal eyes	Caught 66% of ROP eyes, against the 90% target set on CV